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leaves by causing loss of chlorophyll pigment decreasing the rate of photosynthesis and changing the rate of proteins and nucleic acid synthesis.

8. Agricultural Role

- In *Cannabis sativa*, induces male flower formation on female plants.
- Induction of flowers in short day plants.
- It promotes sprouting in storage organs like Potato.
- ABA plays an important role in plants during water stress drought conditions.
- It inhibits the shoot growth and promotes growth of root system. This character protect the plants from water stress. Hence, ABA is called as **stress hormone.**

15.3 Plant Movements



Plants have the capacity for changing their positions in response to external or internal stimuli, which are known as **plant movements**. Movements are basically of two types: I. Vital movements and II. Physical movements (hygroscopic) (Figure 15.14).

I. Vital movements

Vital movements are those which are exhibited by the living cells or plants or organs and they are always related to the irritability of the protoplasm. These movements are of two types:

A. Movements of locomotion

B. Movements of curvature

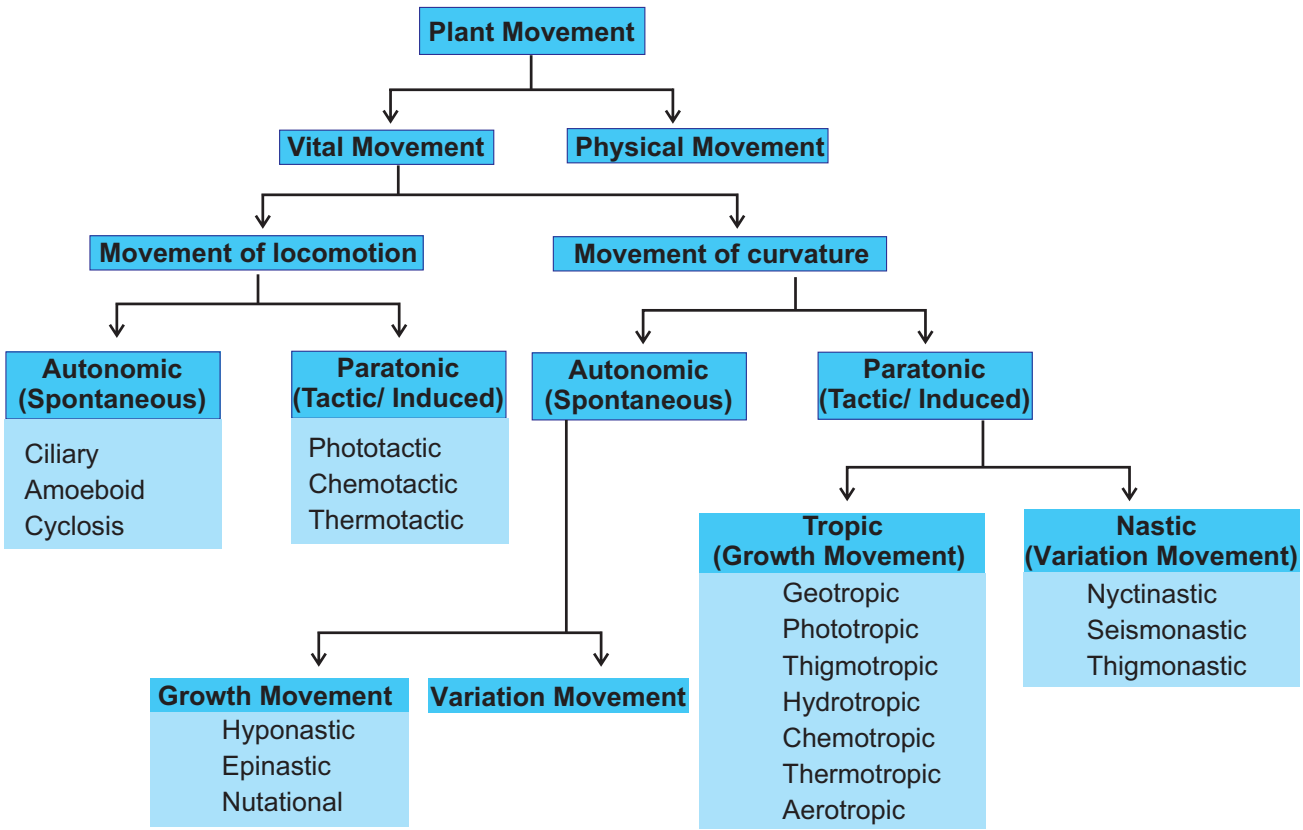


Figure 15.14: Types of Plant Movements